

Association between Interictal Spike Rate and Seizure Frequency in a Large Epilepsy Cohort

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ABSTRACT

Importance: Tracking and predicting seizure frequency in patients with epilepsy is important for prognostication and therapy management. Interictal spikes have been proposed as a biomarker of seizure burden, but their association with seizure frequency has not been well quantified across epilepsy subtypes.

Objective: To measure the association between spike rate and seizure frequency and how this varies by epilepsy subtype.

Design, Setting and Participants: We studied 3,614 consecutive routine outpatient EEGs from 3,245 patients with epilepsy. A validated automated detector (SpikeNet2) estimated spike frequency. Validated large language models performed natural language processing on outpatient clinic notes to extract seizure frequency and epilepsy subtype.

Main Outcomes and Measures: Spearman correlation between spike frequency (spikes/hour) and seizure frequency (seizures/month) for all patients with epilepsy and for patients with generalized epilepsy, temporal lobe epilepsy, and frontal lobe epilepsy.

Results: Overall, spike frequency was modestly associated with seizure frequency ($N = 3,245$, $\rho = 0.11$, $p < 0.001$). Significant positive associations were observed in generalized epilepsy ($N = 625$, $\rho = 0.23$, Bonferroni-adjusted $p < 0.001$) and temporal lobe epilepsy ($N = 834$, $\rho = 0.12$, $p = 0.0013$), but not in frontal lobe epilepsy ($N = 263$, $\rho = 0.11$, $p = 0.22$).

Conclusions and Relevance: In this large outpatient cohort, higher interictal spike rates on routine EEG were associated with higher seizure frequencies, with the strongest relationship observed in generalized epilepsy. These associations support interictal spike rate as a quantitative EEG marker of seizure burden. Spike rate may have clinical utility for risk stratification at diagnosis and for monitoring longitudinal changes in seizure burden in response to therapy.

INTRODUCTION

Epilepsy is characterized by recurrent unprovoked seizures. Accurate estimation of seizure frequency informs counseling, monitoring intensity, and therapy modifications. However, reliable biomarkers of seizure frequency are lacking, and current practice requires weeks to months of observation before seizure burden can be assessed.

Interictal spikes are brief paroxysmal EEG waveforms strongly associated with epilepsy diagnosis, but their relationship to seizure frequency remains uncertain, in part due to the absence of scalable methods to quantify spikes and seizure outcomes in large cohorts.¹⁻⁶ Advances in natural language processing (NLP)⁷ and automated spike detection⁸ now allow estimation of seizure and spike frequency at scale. EEG-based biomarkers are promising because scalp EEG is inexpensive and widely available, and emerging subgaleal EEG devices enable chronic monitoring⁹.

We applied validated large language models (LLM)^{7,10} and an automated spike detector⁸ to electronic medical records and routine EEGs from a large epilepsy cohort to quantify the association between spike rate and seizure frequency.

METHODS

We analyzed patients evaluated in the Penn Epilepsy Center between 2014-2024 who underwent routine outpatient EEGs (<4 hours). The study was approved by the University of Pennsylvania IRB with waiver of informed consent. Full methods are described in **Supplementary Materials**.

Estimating epilepsy type and seizure frequency

Outpatient clinic visits were processed using validated LLMs^{7,10} that extracted seizure frequency (seizures/month), epilepsy diagnosis, epilepsy classification (generalized or focal), and focal localization. Patients without an LLM-confirmed epilepsy diagnosis, documented seizure frequency, or EEG were excluded.

Estimating spike rate

We processed outpatient routine EEGs using SpikeNet2⁸, a validated automated detector. Spike rate was defined as detections per hour of EEG. Clinically reported spike presence was extracted from EEG reports.

Correlating spike rate with seizure frequency

Primary analyses examined the association between mean patient-level spike rate and mean seizure frequency, with secondary analyses stratified by epilepsy subtype. To assess whether spike rate tracks seizure burden over time within individuals, we also tested whether spike-seizure correlations were stronger for clinic visits temporally closer to EEG recordings.

Statistical analyses

Analyses were conducted in MATLAB R2024a. Confidence intervals were estimated via bootstrapping (5,000 iterations). Group comparisons used nonparametric tests with Bonferroni correction for subgroup analyses. Associations were assessed using Spearman correlation. Code is publicly available at https://github.com/erinconrad/seizure_severity.

RESULTS

Cohort summary

We included 3,245 patients (3,614 EEGs). Median [95% CI] monthly seizure frequency was 0.73 [0.62-0.92], and median [95% CI] spikes/hour across EEGs was 1.79 [1.69-1.92] (Table 1).

Spike rates by patient groups

Automatically detected spike rates were higher in EEGs with clinically-reported spikes (median spike rate 10.39 [95% CI 7.83-11.39] spikes/hour) than in EEGs without reported spikes (1.46 [1.37-1.60] spikes/hour) ($p < 0.001$, Cliff's $\delta = 0.65$; Fig. 1A), suggesting a low false-positive detection rate. Spike rates differed across

epilepsy subtypes (Kruskal–Wallis $p < 0.001$, $\eta^2 \approx 0.014$), with higher rates in generalized epilepsy than temporal or frontal lobe epilepsy (Fig. 1B).

Relationship between spike rate and seizure frequency

Spike rate and seizure frequency were positively correlated ($N = 3,245$, $\rho = 0.11$ [95% CI 0.07-0.14], $p < 0.001$). Subtype-specific correlations were significant for generalized epilepsy ($N = 625$, $\rho = 0.23$ [0.15-0.30], Bonferroni-adjusted $p < 0.001$) and temporal lobe epilepsy ($N = 834$, $\rho = 0.12$ [0.05-0.19], $p = 0.0013$), but not frontal lobe epilepsy ($N = 263$, $\rho = 0.11$ [-0.02-0.24], $p = 0.22$; Fig 2A-D). Results were similar when restricting analyses to patients with detectable spikes, although the correlation for frontal lobe epilepsy was larger and significant in this secondary analysis (Fig. S1). Patients with spikes clinically-reported on at least one EEG also had higher mean seizure frequencies than those without spikes (Fig. S2). Spike-seizure correlations were stronger when clinic visits occurred closer in time to EEG acquisition (Fig. S3), suggesting that spike rates may track seizure burden over time.

DISCUSSION

In this large outpatient cohort, higher interictal spike rates were associated with higher seizure frequencies, with the strongest associations in generalized epilepsy. These findings suggest that the spike burden reflects disease severity and may serve as an accessible biomarker of seizure risk.

Prior studies have reported mixed results regarding spike-seizure associations^{1–4}, but the largest studies found that higher spike burden predicts increased seizure frequency or recurrence risk^{5,6}. The stronger associations observed in generalized epilepsy may reflect greater biological relevance or improved detectability of generalized spikes¹¹. The significant associations we observed in temporal lobe epilepsy—and in frontal lobe epilepsy when restricting to patients with detectable spikes—suggest that spike burden also reflects seizure severity in focal epilepsy.

Although the observed correlations are modest and unlikely to directly influence current clinical decision-making, they provide strong evidence that a relationship between spike burden and seizure frequency exists. A supplementary analysis revealed stronger spike–seizure correlations when clinic visits occurred closer in time to EEG acquisition, consistent with the hypothesis that spike rates track seizure frequency over time. However, this analysis does not constitute a true within-patient longitudinal comparison, as too few patients had repeated EEGs closely aligned to clinic visits. In contrast, chronic intracranial EEG studies demonstrate that reductions in spike rates track improvements in seizure control, but these studies are largely restricted to patients with drug-resistant focal epilepsy^{12,13}. Together, these findings underscore the need for chronic scalp or subgaleal EEG studies to determine whether interictal EEG biomarkers can track time-varying seizure burden and guide treatment response.

Limitations

Automated estimation of seizure frequency and spike burden enabled analysis of a cohort an order of magnitude larger than prior studies, but introduces noise, as reflected by non-zero spike rates in some EEGs without clinically reported spikes. Also, we lacked detailed medication administration data to test whether certain medications influenced spike rates.

Conclusions

Higher interictal spike rates on routine outpatient EEG were associated with higher seizure frequencies. These results support spike burden as a promising noninvasive biomarker of seizure severity and motivate future chronic EEG studies to enable individualized epilepsy management.

CONFLICTS OF INTEREST

MBW is a co-founder, serves as a scientific advisor and consultant to, and has a personal equity interest in Beacon Biosignals.

FUNDING

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TABLES AND FIGURE LEGENDS

Patients	3245
Age at first visit	
Median (IQR)	35.6 (25.1-52.6)
Sex	
N (%)	
Female	1750 (53.9%)
Male	1494 (46.0%)
Unknown/Other	1 (0.0%)
Epilepsy subtype	
N (% Epilepsy)	
Temporal lobe	834 (25.7%)
Frontal lobe	263 (8.1%)
Generalized	625 (19.3%)
Other	1175 (36.2%)
Unknown	348 (10.7%)
Number of clinic visits	
Median (IQR)	5.0 (2.0-10.0)
Number of EEGs	
Median (IQR)	1.0 (1.0-1.0)
Mean monthly seizures across visits	0.73 (0.00-5.07);
Median (IQR)	median CI [0.62-0.92]
Mean spikes/hour across EEGs	1.79 (0.64-4.79);
Median (IQR)	median CI [1.69-1.93]
Reported spikes	
N (% EEGs)	
Present	375 (10.4%)
Absent	2683 (74.2%)
Unknown	556 (15.4%)

Table 1. Cohort characteristics. Seizure frequency reflects the mean number of reported seizures per month across clinical visits. Spike rate represents the mean automated spike-detection rate across outpatient routine EEGs. Epilepsy diagnosis, epilepsy subtype, and mean monthly seizure frequency are variables extracted from the large language models. Age, sex, number of clinic visits, and the presence or absence of reported spikes were structured variables extracted from the electronic medical record.

Figure 1. Spike rates across EEGs and cohorts. **A:** EEG-level spike rates detected by SpikeNet2 (spikes/hour, $\log_{10}$ scale) for outpatient routine EEGs stratified by whether the clinical EEG report indicated “present” or “absent” epileptiform discharges. Red points represent individual EEGs, overlaid on boxplots (boxes indicate the interquartile range (IQR); center lines indicate the median; whiskers extend to $1.5 \times \text{IQR}$). Black circles and vertical lines overlying the median represent 95% confidence intervals for the median derived from bootstrapping (5,000 iterations). The horizontal dotted line at 0 denotes EEG with a detected spike rate of 0, arbitrarily offset on the log scale from EEGs with non-zero spike rates. Most EEGs were less than one hour, and so spike rates between 0 and 1 per hour were not possible for most EEGs. For 556/3,614 EEGs, we could not automatically determine the presence or absence of spikes from the report. **B:** Per-patient mean SpikeNet2 spike rates (spikes/hour, $\log_{10}$ scale) stratified by epilepsy subtype (generalized, temporal lobe, frontal lobe). A Kruskal–Wallis test showed overall group differences, with Bonferroni-corrected pairwise Mann–Whitney U-tests shown above comparisons. *** = Bonferroni-corrected $p < 0.001$. ns = Bonferroni-corrected $p > 0.05$. Generalized epilepsy demonstrated higher spikes/hour (median 2.65 95% CI [2.26-3.15]) than temporal (1.89 [1.66-2.16]; Bonferroni-adjusted $p < 0.001$) and frontal (1.62 [1.30-1.97]; $p < 0.001$). The temporal versus frontal comparison was not significant ($p = 1.00$).

Figure 2. Relationship between spike rate and seizure frequency. **A–D:** Scatterplots showing the relationship between mean seizure frequency (seizures/month, $\log_{10}$ scale) and mean spike rate (spikes/hour, $\log_{10}$ scale). Each panel shows individual patients, fitted regression lines (for non-zero points), and Spearman correlation coefficients with associated 95% confidence intervals (obtained via Bootstrapping) and p-values (Bonferroni-corrected for subtype panels). The horizontal line at 0 spikes/hour with asterisks indicates patients with a mean spike rate of 0, and the vertical line at 0 seizures/month with asterisks indicates patients with a mean seizure frequency of 0 (arbitrarily offset from the rest of the data on the log scale). **A** shows results for all patients with epilepsy, **B** shows results for frontal lobe epilepsy, **C** shows results for temporal lobe epilepsy, and **D** shows results for generalized epilepsy.

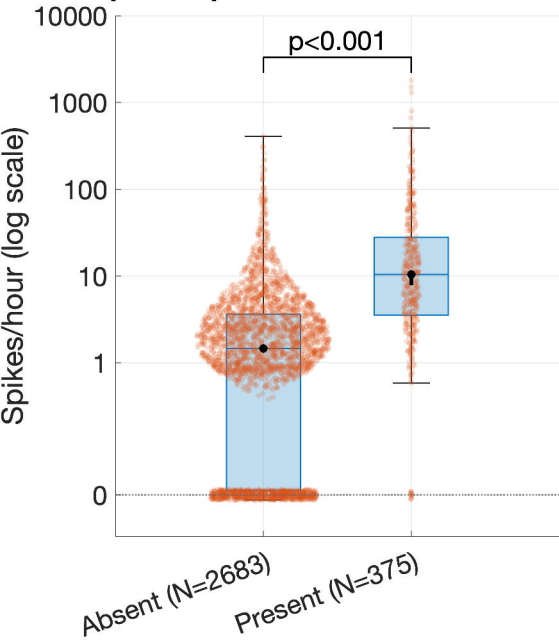
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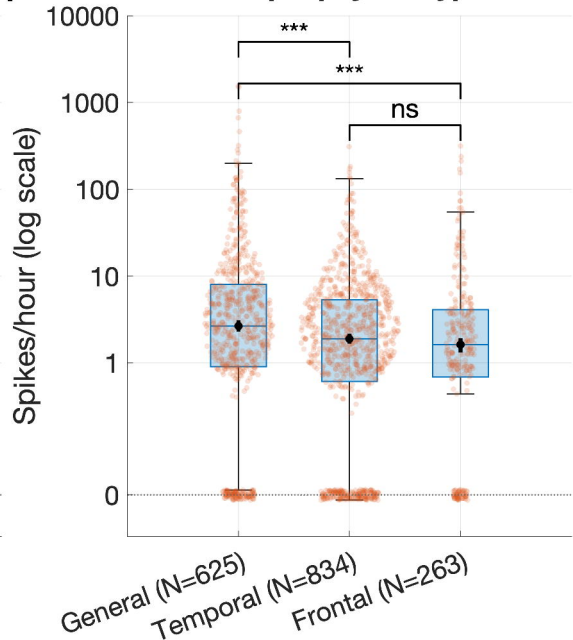
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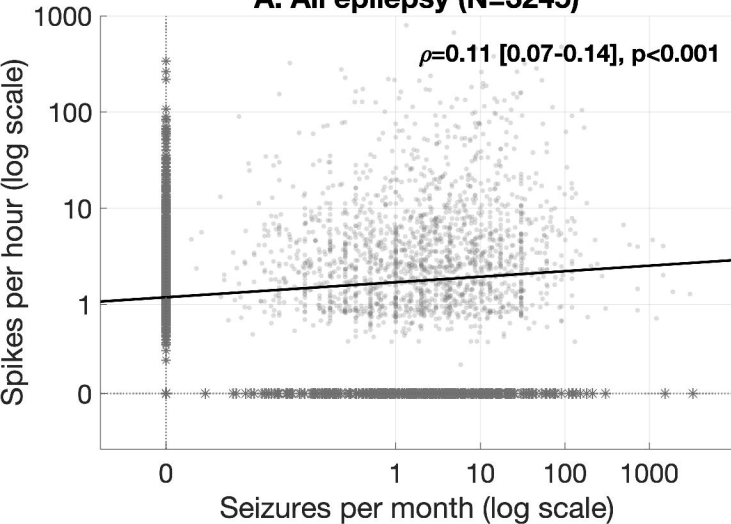
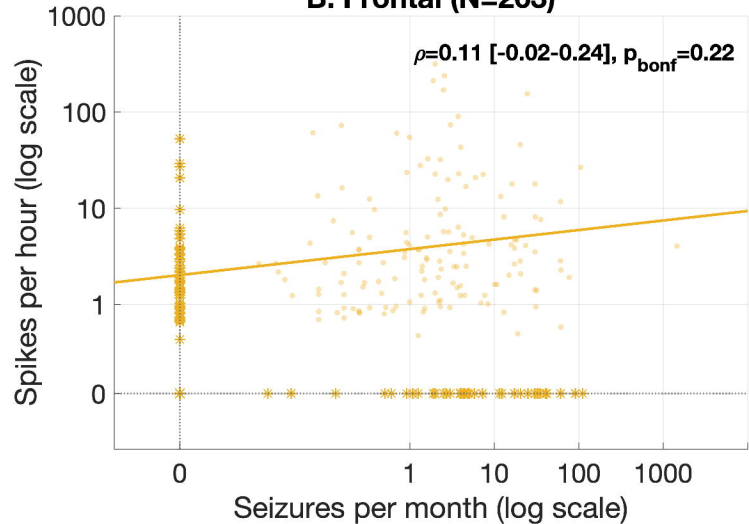
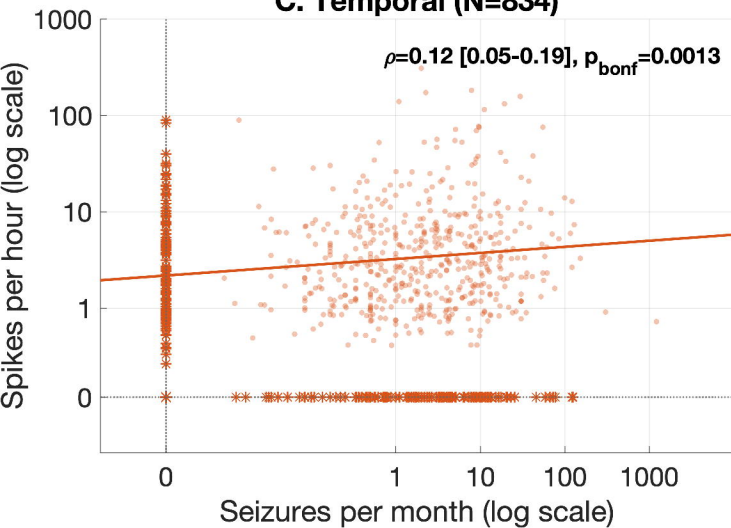
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A. Reported presence or absence of spikes



B. Epilepsy subtype



A. All epilepsy (N=3245)**B. Frontal (N=263)****C. Temporal (N=834)****D. General (N=625)**